Matrices and Determinant

January 5, 2026

Definition of a Matrix

A **matrix** is a rectangular arrangement of numbers in rows and columns.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

This matrix has 2 rows and 3 columns and is of order 2×3.

Basic Operations on Matrices

To add or subtract, two matrices must be of the same order. The sum or difference is determined by adding or subtracting corresponding elements. For example;

Addition

Example 1: $\begin{pmatrix} 2 & 9 & 3 \\ 1 & 5 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 2 & -3 \\ 3 & 7 & 4 \end{pmatrix}$

$$= \begin{pmatrix} 2+1 & 9+2 & 3+(-3) \\ 1+3 & 5+7 & -2+4 \end{pmatrix} = \begin{pmatrix} 3 & 11 & 0 \\ 4 & 12 & 2 \end{pmatrix}$$

Example 2: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$

Basic Operations on Matrices(**Subtraction**)

Example 1:

$$\begin{pmatrix} -1 & 9 \\ 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 3 & 7 \\ 1 & -4 \end{pmatrix}$$

$$\begin{pmatrix} -1-1 & 9-2 \\ 1-3 & 2-7 \\ 3-2 & 4-(-4) \end{pmatrix} = \begin{pmatrix} -2 & 7 \\ -2 & -5 \\ 2 & 8 \end{pmatrix}$$

Example 2:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix}$$

Types of Matrices (Overview)

Common types of matrices include:

- Row matrix
- Column matrix
- Rectangular matrix
- Square matrix
- Zero matrix
- Diagonal matrix
- Identity matrix
- Symmetric matrix
- Skew-symmetric matrix
- Triangular matrix

Row and Column Matrices

Row Matrix: A matrix with only one row.

$$\begin{bmatrix} 2 & 5 & -1 \end{bmatrix}$$

Column Matrix: A matrix with only one column.

$$\begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$$

Rectangular and Square Matrices

Rectangular Matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Square Matrix:

$$\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$$

Zero and Diagonal Matrices

Zero Matrix:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Diagonal Matrix:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Identity Matrix

An identity matrix is a diagonal matrix with 1s on the main diagonal.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Symmetric Matrix

A square matrix is symmetric if:

$$A^T = A$$

Example:

$$\begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$$

Skew-Symmetric Matrix

A square matrix is skew-symmetric if:

$$A^T = -A$$

Properties:

- All diagonal elements are zero

Example:

$$\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Triangular matrix

This is a square matrix in which elements below and/or above the principal diagonal are all zeros. We have mainly two types of triangular matrices.

- A square matrix whose all elements above the main diagonal are zero is called a **lower triangular matrix**. That is, $a_{ij} = 0$ when $i < j$ and $a_{ij} \neq 0$ when $i > j$. For example

$$\begin{pmatrix} 2 & 0 & 0 \\ 3 & 3 & 0 \\ 1 & 4 & 1 \end{pmatrix}$$

- A square matrix whose all elements below the main diagonal are zero is called an **upper triangular matrix**. That is, $a_{ij} = 0$ when $i > j$ and $a_{ij} \neq 0$ when $i < j$.

$$\begin{pmatrix} 3 & 1 & -3 \\ 0 & 2 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

Minor of an Element

The **minor** of an element is obtained by deleting:

- its row
- its column

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Minor of 2 is 3.

Transpose of a Matrix

The **transpose** of a matrix A denoted by A^T is obtained when rows of matrix A are converted to columns.
Example: If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Then

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Matrix Multiplication

Matrix multiplication is defined only when:

$$\text{Columns of first matrix} = \text{Rows of second matrix}$$

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

Scalar Multiplication

Multiplying a matrix by a number multiplies every entry.

Example:

$$3\begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ 0 & 12 \end{pmatrix}$$

Example: Matrix Multiplication

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix}$$

$$AB = \begin{pmatrix} (1 \cdot 2 + 2 \cdot 1) & (1 \cdot 0 + 2 \cdot 5) \\ (3 \cdot 2 + 4 \cdot 1) & (3 \cdot 0 + 4 \cdot 5) \end{pmatrix} = \begin{pmatrix} 4 & 10 \\ 10 & 20 \end{pmatrix}$$

Important Note on Multiplication

Matrix multiplication is **not commutative**:

$$AB \neq BA$$

Students must always check the order before multiplying matrices.

Determinant of a 2×2 Matrix

For:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$|A| = ad - bc$$

If $|A| = 0$, then A has no inverse.

Determinant of a 2×2 Matrix

For:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$|A| = ad - bc$$

Example:

$$\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \Rightarrow |A| = 12 - 2 = 10$$

Determinant of a 3×3 Matrix

For:

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Example: 3×3 Determinant

Find the determinant of:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$|A| = 1(45 - 48) - 2(36 - 42) + 3(32 - 35)$$

$$= -3 + 12 - 9 = 0$$

Properties of Determinant of a Matrix

1. Determinant evaluated across any row or column is the same.
2. The determinant of a null matrix is zero (0).
3. The determinant of an identity matrix is 1.
4. The determinant of a square matrix has the value as that of the determinant of the transposed matrix.
5. If any two row (or two column) of a determinant are interchanging the value of the determinant is multiplied by -1.
6. Let A and B be two matrices of the same size, then

$$\det(AB) = \det(A) \times \det(B).$$

7. Determinant of inverse of a matrix can be defined as

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Cofactor

Cofactor = Minor $\times$ Sign

Steps:

1. Cancel the row and column
2. Find what remains (minor)
3. Apply the sign

Sign Pattern for Cofactors

Sign Pattern for Cofactors of a 2×2 Matrix

$$\begin{bmatrix} + & - \\ - & + \end{bmatrix}$$

Sign Pattern for Cofactors of a 3×3 Matrix

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

The sign depends only on the position of the element.

Cofactor Example (2×2)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Cofactors:

$$C_{11} = d, \quad C_{12} = -c$$
$$C_{21} = -b, \quad C_{22} = a$$

which is

$$A = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

Adjoint of a Matrix

The **adjoint** of a matrix is the transpose of the cofactor matrix.

$$\text{adj}(A) = (\text{cofactor matrix})^T$$

Adjoint of a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Inverse of a Matrix

A matrix A has an inverse A^{-1} if:

$$AA^{-1} = A^{-1}A = I$$

Only **square matrices** with non-zero determinant have inverses.

Inverse of a 2×2 Matrix

If $|A| \neq 0$, then:

$$A^{-1} = \frac{1}{\det(A)} adj\ A$$

Example: Inverse of a Matrix

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$$

$$|A| = (2)(3) - (1)(5) = 1, \text{cofactor} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \text{ and } \quad adj\ A = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$$

Example 2: Inverse of a Matrix

Find the inverse of the matrix

$$\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}.$$

Solution

The determinant is $3 \times 4 - 2 \times 1 = 12 - 2 = 10$.

The adjoint is

$$\begin{pmatrix} 4 & -2 \\ -1 & 3 \end{pmatrix}.$$

So, the inverse is

$$\frac{1}{10}\begin{pmatrix} 4 & -2 \\ -1 & 3 \end{pmatrix}.$$